USER SCENARIOS

Smart Waste Management System

Scenario 1: Bin Monitoring and Overflow Detection

Scenario: A waste operations manager wants to identify bins that are close to overflow and prioritize them for collection.

Main Sequence:

1. The manager logs into the Smart Waste Management System dashboard.
2. The manager opens the Bin Monitoring module.
3. The system displays simulated bin fill levels, bin type, location, and operational status on the map.
4. The manager filters the view to show bins that have exceeded the alert threshold.
5. The system classifies the alerts by severity and highlights high-priority bins with visual indicators.
6. The manager acknowledges the urgent alerts and reviews the affected zones.
7. The manager selects the bins that require urgent attention.
8. The system stores the selected bins as collection priorities for route planning.

Alternative Sequence:

- If bin data is temporarily unavailable, the system displays the last recorded value and marks the bin for refresh.
- If a bin is marked as damaged, the system flags it for maintenance instead of regular collection.

Special Requirements:

- Real-time or simulated fill-level updates.
- Color-coded map visualization for bin status.
- Threshold-based alert detection and severity classification.

Post-condition: The manager has an updated list of high-priority bins for operational planning.

Scenario 2: Bin Registration for a New Service Zone

Scenario: An authorized administrator wants to register a new bin for a newly covered service area.

Main Sequence:

1. The administrator logs into the system with authorized access.
2. The administrator opens the Bin Management module.
3. The administrator selects the option to register a new bin.
4. The system displays a form for unique bin ID, type, capacity, location, and assigned service zone.
5. The administrator enters the required information and confirms the registration.
6. The system validates the entered data and stores the new bin record.
7. The system makes the bin available in map views and operational lists.

Alternative Sequence:

- If the entered bin ID already exists, the system prompts the administrator to provide a unique ID.
- If required fields are missing, the system prevents submission until the missing information is completed.

Special Requirements:

- Role-based access for bin registration.
- Validation of unique identifiers and required fields.
- Immediate visibility of newly registered bins in the operational map.

Post-condition: A new bin is registered successfully and becomes part of the operational system.

Scenario 3: Bin Information Update and Deactivation

Scenario: An authorized manager wants to update a bin's details or deactivate a bin that is no longer in service.

Main Sequence:

1. The manager logs into the system.
2. The manager opens the Bin Management module and searches for a specific bin.
3. The system displays the current bin details, status, and operational history.
4. The manager updates the bin information such as location, type, capacity, or status.

5. If the bin is no longer in use, the manager selects the deactivate option.

6. The system saves the change and records it in the bin history.

7. The updated or deactivated bin record is reflected in future operational views and reports.

Alternative Sequence:

• If the manager enters invalid location or capacity data, the system requests correction before saving.

• If the bin is linked to an active route, the system warns the manager before deactivation.

Special Requirements:

• Only authorized users may update or deactivate bins.

• Historical tracking must be preserved after each change.

• Deactivated bins must not appear in active collection planning.

Post-condition: The selected bin is updated or deactivated, and the change is stored in the historical record.

Scenario 4: Route Optimization for Daily Collection

Scenario: A manager wants to generate an efficient waste collection route based on current bin status and operational constraints.

Main Sequence:

1. The manager accesses the Route Optimization module.

2. The system retrieves the current list of bins requiring collection.

3. The manager selects the collection zone, available truck, and driver assignment.

4. The system checks truck availability and driver eligibility for the selected route.

5. The manager enters route constraints such as truck capacity and shift duration.

6. The system calculates an optimized collection route based on priority bins, location, and operational constraints.

7. The optimized route is displayed on the map with stop order and estimated completion time.

8. The manager reviews the generated route and confirms it for execution.

9. The system creates the route schedule and assigns the route to the selected driver.

Alternative Sequence:

- If the selected truck does not have enough capacity, the system suggests a different truck or split route.
- If route optimization cannot be completed due to missing data, the system notifies the manager and allows manual adjustment.

Special Requirements:

- Fast route calculation for operational use.
- Support for truck capacity, zone boundaries, and shift constraints.
- Map-based route visualization and scheduling support.

Post-condition: An optimized and approved route is scheduled and assigned for waste collection.

Scenario 5: Driver Executes Assigned Route

Scenario: A driver wants to complete daily collection tasks using the route assigned by the system.

Main Sequence:

1. The driver logs into the driver interface on a mobile device.
2. The system displays the assigned route, stop sequence, and list of bins to be collected.
3. The driver starts the shift and activates route navigation.
4. The system marks the route as in progress.
5. The driver travels to each stop and performs collection activities.
6. After servicing a bin, the driver confirms the stop as completed in the system.
7. If a collection stop cannot be completed, the driver records the missed collection and the reason.
8. The system updates route progress and collection status in real time.
9. The driver finishes the route and submits the shift summary.

Alternative Sequence:

- If road access is blocked, the driver reports the issue through the incident reporting function.
- If internet connectivity is lost, the driver continues working offline and the data is synchronized when the connection is restored.

Special Requirements:

- Mobile-friendly interface for drivers.
- Offline support for field operations.
- Step-by-step route and stop management.

Post-condition: The assigned collection route is completed and recorded in the system.

Scenario 6: Fleet Tracking and Vehicle Status Monitoring

Scenario: A fleet manager wants to track trucks and monitor their operational status during collection activities.

Main Sequence:

1. The fleet manager opens the Fleet Management dashboard.
2. The system displays all active trucks with simulated location, status, assigned driver, and route information.
3. The fleet manager selects a specific truck to view detailed information.
4. The system shows fuel usage, availability status, and route progress for the selected vehicle.
5. The fleet manager compares active vehicles across zones and identifies delays or issues.
6. The system updates truck status continuously during working hours.
7. The fleet manager uses the displayed information to coordinate fleet operations.

Alternative Sequence:

- If the vehicle location feed is interrupted, the system displays the last known location and status timestamp.
- If a truck reports a breakdown, the system alerts the manager and marks the vehicle as unavailable.

Special Requirements:

- Real-time or near real-time vehicle visibility.
- Truck registration data must be available in the fleet dashboard.
- Support for fuel and availability indicators.

Post-condition: The fleet manager has an updated view of vehicle operations and can intervene when needed.

Scenario 7: Vehicle Maintenance Management

Scenario: A maintenance coordinator wants to record and manage truck servicing activities.

Main Sequence:

1. The maintenance coordinator logs into the Fleet Management module.
2. The coordinator opens the maintenance records for a selected truck.
3. The system displays service history, current condition, and upcoming maintenance schedule.
4. The coordinator records a new maintenance event or updates the vehicle condition.
5. The system stores the maintenance information and recalculates the next service date.
6. If the vehicle is unavailable for operations, the system marks it as under maintenance.
7. The updated maintenance status is reflected in fleet planning views.

Alternative Sequence:

- If a scheduled service is overdue, the system generates a maintenance alert.
- If maintenance data is incomplete, the system prompts the coordinator to provide the missing details.

Special Requirements:

- Maintenance history tracking.
- Automatic service reminders and alerts.
- Secure update of operational vehicle status.

Post-condition: Vehicle maintenance information is updated and reflected in fleet planning.

Scenario 8: Driver Assignment and Shift Preparation

Scenario: A manager wants to assign drivers to trucks and prepare them for daily route execution.

Main Sequence:

1. The manager logs into the operational planning module.
2. The system displays available drivers, their shift status, and assignment eligibility.
3. The manager selects a truck that is available for collection.
4. The manager assigns an eligible driver to the truck and route.
5. The system updates the assignment and sends task information to the driver interface.
6. The driver reviews the assigned route, stop sequence, and task updates before starting the shift.

Alternative Sequence:

- If a selected driver is unavailable, the system prevents the assignment and suggests another eligible driver.
- If the selected truck is not operational, the system requests the manager to choose another vehicle.

Special Requirements:

- Driver availability and assignment rules must be enforced.
- Task updates must be visible on the mobile interface before route start.

Post-condition: The driver, truck, and route are linked correctly for operational execution.

Scenario 9: Citizen Reports a Waste-Related Issue

Scenario: A citizen wants to report an overflowing bin or illegal dumping through the platform.

Main Sequence:

1. The citizen opens the Citizen Reporting interface.
2. The citizen enters the issue location using map selection or device location.
3. The citizen selects the issue type and writes a short description.
4. The citizen uploads a photo of the issue.
5. The system validates the submitted information.
6. The citizen submits the report.
7. The system creates a new report entry and assigns it a tracking status.
8. The system forwards the report for operational handling.

Alternative Sequence:

- If the photo upload fails, the system allows the citizen to retry or submit the report without the image.
- If the location is incomplete, the system prompts the citizen to provide a valid location before submission.

Special Requirements:

- Simple reporting workflow for non-technical users.
- Secure handling of citizen data and uploaded media.
- Validation of required report information before submission.

Post-condition: The citizen’s report is successfully recorded and forwarded for action.

Scenario 10: Manager Assigns and Tracks Citizen Reports

Scenario: A manager wants to review incoming citizen reports and assign them to the appropriate field staff.

Main Sequence:

1. The manager logs into the Citizen Reporting module.
2. The system displays a list of new and active reports with location, priority, and status.
3. The manager opens a selected report to review its details and attached photo.
4. The manager assigns the report to a field operator or links it to an active collection route.
5. The system updates the report status to indicate that it is in progress.
6. After the issue is resolved, the manager or operator updates the report status to resolved.
7. The system stores the full status history of the report.

Alternative Sequence:

- If the report is duplicated, the system allows the manager to merge it with an existing case.
- If no operator is available, the system keeps the report pending and highlights it for follow-up.

Special Requirements:

- Status tracking throughout the report lifecycle.
- Role-based assignment and update permissions.
- Audit trail for issue handling.

Post-condition: Citizen complaints are assigned, tracked, and resolved through the system.

Scenario 11: Citizen Views Report Status

Scenario: A citizen wants to check the current status of a previously submitted waste-related report.

Main Sequence:

1. The citizen opens the Citizen Reporting interface.
2. The citizen navigates to the submitted reports section.
3. The system displays the list of reports associated with the citizen account or tracking reference.
4. The citizen selects a report to view its current status.
5. The system shows the report status, assigned handler, and latest update information.

Alternative Sequence:

- If the tracking reference is invalid, the system informs the citizen and requests a valid reference.
- If the report has already been resolved, the system shows the closure details.

Special Requirements:

- Clear status visibility for citizens.
- Simple and transparent access to report updates.

Post-condition: The citizen can see the latest handling status of the submitted report.

Scenario 12: Recycling Monitoring and Contamination Control

Scenario: A recycling supervisor wants to monitor recycling bins and detect contamination issues.

Main Sequence:

1. The supervisor opens the Recycling Management module.
2. The system displays recycling bin locations, fill levels, and contamination alerts.
3. The supervisor reviews bins flagged as contaminated or improperly used.
4. The system shows recent collection records and contamination history for the selected recycling bin.
5. The supervisor decides whether to schedule collection, inspection, or awareness action.
6. The system records the supervisor’s decision and updates the recycling status.

Alternative Sequence:

- If contamination data is unavailable, the system marks the bin for manual inspection.
- If contamination exceeds the acceptable level, the system generates a high-priority alert.

Special Requirements:

- Separate handling of recycling operations from general waste.
- Support for contamination alerts and recycling performance tracking.
- Historical record of recycling bin activity.

Post-condition: Recycling bins are monitored properly and contamination issues are identified for action.

Scenario 13: Reporting and Analytics for Operational Decisions

Scenario: An operations manager wants to generate reports and review KPIs for waste collection performance.

Main Sequence:

1. The manager opens the Reporting and Analytics module.
2. The system displays available report types and KPI dashboards.
3. The manager selects a report period, zone, or operational category.
4. The system generates charts, summaries, and performance indicators such as collected bins, response time, truck utilization, and recycling rate.
5. The manager reviews the dashboard and compares trends across zones.
6. The manager exports the report in PDF or CSV format for sharing or record-keeping.

Alternative Sequence:

- If the selected dataset is incomplete, the system notifies the manager and marks missing values clearly.
- If export fails, the system allows the user to retry the export process.

Special Requirements:

- Accurate KPI calculation and trend visualization.
- Support for PDF and CSV export.
- Near real-time analytics refresh.

Post-condition: The manager receives a clear operational report for decision-making and municipal reporting.

Scenario 14: System Handles Dynamic Route Changes During Operations

Scenario: The system needs to react when new urgent bins or incidents appear while collection is already in progress.

Main Sequence:

1. The system continuously monitors active routes, new alerts, and incoming citizen reports.
2. A new urgent event is detected, such as an overflowing bin or a high-priority complaint.
3. The system evaluates the location of the active trucks and the urgency of the new task.
4. The system recommends a route adjustment for the most suitable driver or team.
5. The manager reviews the suggested route change and approves it.
6. The updated route is sent to the driver's mobile interface.
7. The driver receives the change and continues collection using the revised route.

Alternative Sequence:

• If the manager rejects the proposed change, the original route remains active and the new task is queued separately.

• If no available truck can handle the urgent task, the system escalates the issue and marks it for manual scheduling.

Special Requirements:

• Continuous event monitoring during operations.

• Dynamic route recalculation based on operational priorities.

• Immediate communication of route updates to field staff.

Post-condition: Urgent operational events are integrated into live collection activities with minimal disruption.

Scenario 15: Alert Acknowledgment and Escalation

Scenario: A manager wants to control how critical operational alerts are acknowledged and escalated.

Main Sequence:

1. The manager opens the alerts panel from the dashboard.
2. The system displays active alerts together with their severity levels.
3. The manager reviews the details of a selected alert.
4. The manager acknowledges the alert to indicate that it is under review.
5. The system tracks how long the alert remains unresolved.
6. If the alert is not resolved within the defined time, the system escalates it to higher-level staff.

Alternative Sequence:

- If the alert is resolved before the escalation deadline, the system closes the alert without escalation.
- If multiple alerts occur in the same zone, the system groups them by severity for easier handling.

Special Requirements:

- Severity-based alert management.
- Automatic escalation rules for unresolved critical alerts.

Post-condition: The alert is either acknowledged and resolved or escalated according to the defined policy.

Scenario 16: Map Filtering and Operational Search

Scenario: A manager wants to search the map and focus on specific operational elements for quick decision-making.

Main Sequence:

1. The manager opens the map view from the dashboard.
2. The system displays bins, trucks, and service zones with visual status indicators.
3. The manager applies filters by zone, bin status, and truck status.
4. The manager searches for a specific bin or truck identifier.
5. The system narrows the map view to the matching operational items.
6. The manager uses the filtered information to review current operational priorities.

Alternative Sequence:

- If no results match the selected filters, the system informs the manager and keeps the previous view available.

• If a filtered asset becomes inactive, the system updates the map indicators automatically.

Special Requirements:

• Interactive map visualization.

• Fast search and filtering for operational use.

Post-condition: The manager obtains a focused and searchable operational view of the city service map.

Scenario 17: Truck Registration and Operational Update

Scenario: An authorized fleet administrator wants to register a new truck or update the details of an existing truck.

Main Sequence:

1. The administrator opens the Fleet Management module.
2. The administrator selects the option to register a new truck or update an existing one.
3. The system displays the truck form including vehicle ID, capacity, type, and operational status.
4. The administrator enters or modifies the truck information.
5. The system validates the data and stores the changes.
6. The truck record becomes available for tracking, planning, and route assignment.

Alternative Sequence:

• If the vehicle ID already exists, the system requires a unique identifier before saving.

• If the truck is currently assigned to an active route, the system warns the administrator before major status changes.

Special Requirements:

• Authorized access to fleet records.

• Validation of truck information before storage.

Post-condition: A truck record is created or updated successfully for operational use.

Scenario 18: Bin History Review for Performance Analysis

Scenario: A manager wants to review the historical record of a bin to support operational analysis.

Main Sequence:

1. The manager opens the Bin Management module and selects a specific bin.
2. The system displays the historical record of the bin including fill-level changes, collection events, alerts, and maintenance records.
3. The manager reviews the historical activity to understand recurring overflow or maintenance patterns.
4. The manager compares the bin's performance over time with current operational status.
5. The system keeps the selected history available for future reporting and decision-making.

Alternative Sequence:

- If historical data is incomplete, the system marks missing periods clearly.
- If the selected bin has been deactivated, the system still provides access to its historical record.

Special Requirements:

- Preservation of historical operational records.
- Clear visual presentation of bin history.

Post-condition: The manager gains a complete historical view of the selected bin for analysis and planning.

Scenario 19: Missed Collection Review and Rescheduling

Scenario: A manager wants to review missed collection stops and take corrective action.

Main Sequence:

1. The manager opens the collection monitoring module.
2. The system displays routes with stops marked as delayed, skipped, or missed.
3. The manager selects a route to review missed collection reasons.
4. The system shows the recorded reason for each missed stop.
5. The manager decides whether to reschedule the missed stop or manually override the route plan.
6. The system updates the collection schedule and route execution status accordingly.

Alternative Sequence:

- If no truck is available for immediate rescheduling, the system queues the missed stop for the next available schedule.
- If the missed stop was caused by an incident, the system links it to the recorded field report.

Special Requirements:

- Visibility into missed collection records.
- Support for rescheduling and manual route adjustment.

Post-condition: Missed collection stops are reviewed and incorporated into updated operational planning.

Scenario 20: Secure Login and Role-Based Access

Scenario: A user wants to access the system according to the permissions of their assigned role.

Main Sequence:

1. The user opens the Smart Waste Management System login page.
2. The user enters valid authentication credentials.
3. The system verifies the credentials and the assigned user role.
4. The system grants access only to modules and functions permitted for that role.
5. The user is redirected to the appropriate dashboard, such as manager, driver, administrator, or citizen view.

Alternative Sequence:

- If the credentials are invalid, the system denies access and shows an error message.
- If the user attempts to access a restricted module, the system blocks the action based on role permissions.

Special Requirements:

- Secure authentication for all users.
- Role-based access control for all system modules.

Post-condition: The authenticated user can access only the functions allowed by their role.